PROGRAM

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from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load dataset
digits = load_digits()

X = digits.data
y = digits.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

# Predict digits
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display output
print("Handwritten Digit Recognition")
print("Predicted Digit:", y_pred[0])
print("Actual Digit:", y_test[0])
print("Accuracy:", accuracy * 100, "%")

# Display image
plt.imshow(digits.images[0], cmap='gray')
plt.title("Handwritten Digit")
plt.axis('off')
plt.show()
```